

Solid equilibria and adsorption

After midterm · Session 4 of 6 · 180 minutes

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The question for today

What happens to gas pressure when the solid inventory runs out?

Write your prediction before opening the lab. By the end, support an answer with one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Bulk and surface equilibria

PROBLEM	EQUILIBRIUM DESCRIPTION
SLE	Solute chemical potential equal in liquid and solid
Bulk GSE	Chemical potential equal in gas and solid
Adsorption	Surface loading at gas/solution chemical potential

Surface uptake is not the formation of a macroscopic bulk solid phase.

Sublimation and vessel inventory

At fixed T , a present bulk solid fixes the ideal-gas partial pressure at its sublimation pressure within the model.

In a rigid vessel,

$$n_g^{sat} = \frac{P_{sub}(T)V}{RT}$$

If total subliming-species inventory is below this amount, the solid is exhausted and pressure falls below P_{sub} .

A gas–solid inventory check

Synthetic example: $T=300\text{ K}$, $V=1\text{ L}$, $P_{\text{sub}}=10\text{ kPa}$.

$$n_g^{\text{sat}} = 10 / (8.31446 \times 300) \approx 0.00401\text{ mol}$$

With 0.01 mol total, about 0.00599 mol remains solid. With 0.002 mol total, no solid remains and $P \approx 4.99\text{ kPa}$.

Fixed T, V equilibrium minimizes Helmholtz energy, not Gibbs energy at fixed P .

An independent check

Compare total inventory with $P_{\text{sub}} V/(RT)$. Check gas plus solid moles and distinguish partial pressure from total pressure if an inert gas is present.

One demonstration • 20 minutes

Open the core lab

In Lab 10 choose the gas–solid vessel mode. Keep T and V fixed and vary the subliming-species inventory across the solid-depletion threshold.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 10, compare inventories above and below the solid-depletion boundary at fixed T and V . Check the inventory balance and explain the resulting gas pressure.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

What happens to gas pressure when the solid inventory runs out?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Required practice • 45–60 minutes

In Lab 10, compare inventories above and below the solid-depletion boundary at fixed T and V . Check the inventory balance and explain the resulting gas pressure.

Assessment 4 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is part of this time budget.

Optional study

Choose one extension: compare Henry and Langmuir in Lab 11, check a closed adsorption vessel, or explore pure-solid solubility and the eutectic assumptions in Lab 10.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.